



Review article

Efficacy and safety of perioperative application of esketamine on postpartum depression: A meta-analysis of randomized controlled studies

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ABSTRACT

Postpartum depression (PPD) seriously impairs the physical and mental health of mothers and their offspring, so how to prevent the occurrence of PPD has essential significance. Esketamine is a common general anesthetic that produces rapid and sustained antidepressant effects. However, the efficacy and safety of perioperative esketamine administration for PPD prevention remain uncertain. We conducted a meta-analysis to determine the effect of perioperative intravenous esketamine on PPD. Randomized controlled trials were included. The primary outcome was the prevalence of PPD and postpartum Edinburgh Postnatal Depression Scale (EPDS) scores. Secondary outcomes included postoperative pain scores and esketamine-related adverse effects. Seven studies included 669 patients treated with esketamine and 619 comparisons. Esketamine could effectively reduce EPDS scores and the incidence of PPD after cesarean section. Even at 42 days postpartum, the incidence of PPD was still significantly lower in the esketamine group. Esketamine did not increase the incidence of postoperative nausea and vomiting, dizziness, and drowsiness. In the esketamine low-dose subgroup, postoperative nausea and vomiting were significantly lower in the esketamine group. The two groups had no significant difference in postoperative pain scores. In conclusion, using esketamine during the perioperative period can reduce the incidence of PPD without increasing adverse effects.

1. Introduction

According to the American College of Obstetricians and Gynecologists (ACOG), postpartum depression (PPD) is defined as a major or mild depressive episode that occurs in the first 12 months after delivery (2018). EPDS is the most frequently used and effective screening scale for PPD in the research setting and clinical practice (2018; Vázquez and Míguez, 2019). Six weeks after delivery is a high-risk period for PPD, most occurring within one week (Wang et al., 2022; Di Florio and Meltzer-Brody, 2015). In developed countries, about 13 % of women who give birth experience PPD (Howard et al., 2014). Studies from the Chinese population have reported rates of PPD as high as 10 % to 22 % (Li et al., 2022). PPD damages maternal physical and mental health and seriously affects children's physical and mental health and long-term cognitive development (Slomian et al., 2019). Both prenatal and postnatal depression and anxiety in mothers were associated with poorer social-emotional, cognitive, linguistic, motor, and adaptive behavioral

development in offspring, which extended beyond infancy into childhood and adolescence (Rogers et al., 2020). Maternal and paternal depression was associated with poorer cognitive performance in their offspring six years later, and the severity of maternal depression was correlated with poorer cognitive outcomes (Li et al., 2021). PPD has potentially severe consequences for mothers and their children, and there is an urgent need for safe, effective, and accessible treatments. However, Currently approved antidepressants work slowly, require long-term use, and are less effective (Carlini et al., 2023). Moreover, maternal concerns about the impact of antidepressants on infant breastfeeding have limited the use of antidepressants in maternity (Cox et al., 2016).

Brexanolone is the first and only treatment specifically for PPD and has also been approved by the United States Food and Drug Administration (FDA) (Powell et al., 2020). However, brexanolone is unavailable to most patients due to its high cost (approximately \$34,000) and long infusion time (60 h in an inpatient setting). Ketamine has been used as a

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fast-acting antidepressant. It has shown rapid and significant efficacy in many patients with major depressive disorder (MDD) (Dean et al., 2021) and treatment-resistant depression (TRD) (McIntyre et al., 2021). It has been used with caution because of its psychopathic effects (Shoib et al., 2022).

Esketamine, the mirror isomer of ketamine, has an affinity for N-methyl-D-aspartic acid (NMDA) receptors about 2 to 4 times that of ketamine and has less psychopathic effects (Shoib et al., 2022). On March 5, 2019, the United States FDA approved Johnson & Johnson intranasal esketamine for its use in TRD in combination with an antidepressant (Turner, 2019). Clinical studies on using esketamine to prevent and treat perioperative depression have been increasing (Cheng et al., 2022; Qiu et al., 2022; Chen et al., 2023). The researchers found that esketamine rapidly and effectively reduced the incidence of postoperative depression, postoperative pain, and postoperative opioid use, improved sleep and postoperative recovery quality, and did not increase adverse effects. In a recent study by our research group, we found that esketamine inhibited the production of pro-inflammatory cytokines, effectively treated postoperative depression-like symptoms in missed abortion patients, reduced the dosage of propofol, and reduced the inflammatory response (Jiang et al., 2023). Although the rapid and sustained antidepressant effects of esketamine have been demonstrated in many clinical and basic studies, the safety and efficacy of esketamine in perinatal use remain controversial. Given its potential importance for clinical use, a meta-analysis of the safety and effectiveness of intravenous esketamine in preventing PPD during cesarean section is urgently needed.

2. Methods

This meta-analysis was reported under the Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines (PRISMA) (Moher et al., 2009) and registered in the International Prospective Register of Systematic Reviews (PROSPERO). (Registration number: CRD42023406245).

2.1. Literature search and screening

We used the key words "(esketamine) OR (kataved) OR (s-ketamine OR (S)-2-(o-chlorophenyl)-2-(methylamino)cyclohexanone) OR (L-Ketamine) OR ((-)-Ketamine) OR (spravato)) and ((depression, postpartum) OR (postnatal depression) OR (depression, postnatal) OR (postpartum depression) OR (depression, postpartum) OR (post partum depression) OR (postpartum depression) OR (post-natal depression) OR (depression, post-natal) OR (post natal depression) OR (Edinburgh Postnatal Depression Scale (EPDS))) conducted a systematic literature search and restricted "cesarean section," "written in English or Chinese," "clinical trials" and "randomized controlled trials." The literature search mainly included the Cochrane Central Register of Controlled Trials (CENTRAL), PubMed, Embase and Web of Science databases. The search included dates through April 1, 2023. We considered all potentially eligible studies and conducted a manual search of references for articles that met the inclusion criteria. Literature screening and data extraction were completed independently by two reviewers. Two independent investigators screened the title and abstract in the initial screening stage. The third investigator resolved any disagreements. Subsequently, the full text is screened to determine eligibility for inclusion. The inclusion criteria were as follows: (1) Randomized studies to explore the use of esketamine (experimental group) and treatment as usual (comparison group) in the perioperative prevention of PPD; (2) Articles on human clinical trials of elective cesarean section. The exclusion criteria were as follows: review, retrospective study, meta-analysis, self-controlled studies before and after, case report, case series, conference paper, basic research (including animal experiments), incomplete or unavailable research data, studies where full text is unavailable, and any language other than Chinese or English.

2.2. Data extraction

The extracted data included: (1) basic research information: author name, year of publication; (2) basic information of subjects: sample size, maternal age, height, body weight, body mass index and gestational age; (3) intervention: study design, experimental grouping, anesthesia type, administration timing and dosage of esketamine, treatment of comparison group; (4) results: EPDS score, the threshold of PPD, the incidence of PPD, adverse effects and postoperative pain score. We extracted EPDS scores and PPD incidence from the included studies as primary outcomes. Secondary outcomes were postoperative pain scores and adverse effects (nausea and vomiting, dizziness, drowsiness and other adverse effects). When there is no data in the article, we try to contact the author to ask for the original data or use PlotDigitizer software to get the original data from the figure. Three time points (1–3d, 5–7d, and 42 days after surgery) were selected to evaluate the effect of esketamine. In addition, two trained reviewers independently assessed study quality and risk of bias, according to the Cochrane Collaboration's risk of bias tool (Review Manager 5.4). Any differences were resolved through discussion to reach a consensus.

2.3. Subgroup analysis

To examine whether the effect of esketamine on PPD differed in terms of moderating variables, subgroup analyses were performed based on esketamine administration method (single-dose vs. continuous infusion administration) and esketamine dose (low dose vs. high dose). To further investigate the safety of perioperative use of esketamine as an antidepressant, we performed subgroup analyses of esketamine-associated adverse effects, including esketamine administration methods (single infusion vs. continuous infusion administration through patient-controlled intravenous analgesia (PCIA)) and esketamine dose (low dose vs. high dose).

2.4. Summary of findings

In our review, we used the principles of the GRADE (Grades of Recommendation, Assessment, Development, and Evaluation) system to assess the quality of the body of evidence associated with a particular outcome (incidence of PPD and EPDS scores at 1–3d, 5–7d, and 42d postoperative; incidence of postoperative nausea and vomiting, dizziness, and drowsiness; pain scores at 24 h and 48 h postoperative). We used GRADEprofiler software to build the "Summary of findings" table. The quality of the body of evidence considers the risk of bias within the study, the directness of the evidence, the heterogeneity of the data, the precision of the effect estimates, and the risk of publication bias.

2.5. Statistical analysis

Adopt Review Manager 5.4 and Stata 17.0 software were employed for this meta-analysis. Due to inconsistencies in the scale used to assess PPD and postoperative pain, we chose standardized mean difference (SMD) and 95 % confidence interval (95 % CI) as the effect size (ES) of the continuous variables. We chose risk ratio (RR) and 95 % CI as the ES for dichotomous variables. The significance level was 0.05 in the meta-analysis. In the median and quartiles interval of the report as a result, through online data conversion formula into the mean and standard deviation (math.hkbu.edu.hk/~tongt/papers/median2mean.html). Numerical estimates were obtained using software (Plot Digitizer) when sequential results were represented graphically only. I^2 value was used to estimate inter-study heterogeneity. If the $I^2 < 50\%$, there was no significant heterogeneity. We used the fixed-effect model to calculate the combined ES. If the heterogeneity $I^2 \geq 50\%$, a random effects model was chosen, and we took subgroup analysis and leave-one-out sensitivity analysis to investigate the source of heterogeneity.

3. Results

3.1. Search results and included study characteristics

A total of 125 studies were retrieved, and 34 were included after initial screening. Seven RCTs with 1288 patients were eligible by reading the full text and referencing the inclusion criteria (Zhang et al., 2021; Wang et al., 2022, 2023; Shen et al., 2023; Gu et al., 2021; Gao et al., 2022; Han et al., 2022). The flow chart of the research selection process is shown in Fig. 1, and the features included in this research are shown in Table S1.

3.2. Risk of bias assessment

We used the Cochrane Handbook to assess the risk of bias in included studies (Fig. 2). The explanation is as follows: all the studies reported specific methods for generating random sequences. Three studies reported how allocation concealment was done. Five studies reported particular ways to achieve blindness in participants and researchers. Six studies reported specific methods of achieving blindness in outcome evaluation. The remaining studies that did not mention blind and allocation concealment were assessed as having an "unclear risk of bias". In one study in which researchers were not blinded, the outcome may have been affected by the absence of blindness and was assessed as "high risk". Three studies did not provide a registered research protocol number in their papers and were evaluated as "unclear risk of bias". Although participants lost to follow-up in three studies, the number of participants who dropped out was balanced and did not affect the results. Therefore, we classify this aspect of preliminary outcome studies as a "low risk of

bias". In one study (Han et al., 2022), 300 patients were needed according to the calculation of the sample size of the two groups, and 275 patients were finally included in the study. In the comparison group, 13 cases were lost to follow-up. The lost to follow-up rate was 7.83 %. In the intervention group, 39 people were lost to follow-up, with a lost to follow-up rate of 24.22 %. Therefore, the incomplete outcome bias of this study was rated as "high risk". Financial conflicts of interest were not stated in one study (Gao et al., 2022), and the other biases were rated as "unclear risk of bias".

3.3. Results of the meta-analysis

The first part of our primary meta-analysis investigated the different effects of esketamine and saline on the incidence of PPD from elective cesarean section patients on postoperative days (PODs) 1–3, 5–7 and 42. The antidepressant effect was better in the esketamine group than in the comparison group. Compared with the placebo group, the esketamine group showed positive effects on POD 1–3 (RR=0.42, 95 % CI [0.28, 0.62], $P < 0.0001$, $I^2 = 0$ %), POD 5–7 (RR=0.37, 95 % CI [0.26, 0.53], $P < 0.00001$, $I^2 = 44$ %), and POD 42 (RR=0.33, 95 % CI: [0.14, 0.79], $P = 0.01$, $I^2 = 59$ %). Forest plots of this analysis are presented in Fig. 4.

EPDS scores in the esketamine group were significantly lower than those in the comparison group at 1–3 days postpartum (SMD=−0.62, 95 % CI [−0.82, −0.43], $P < 0.00001$, $I^2 = 0$ %) (Fig. 3A). Fig. 3B shows that there was no significant difference in EPDS scores between the esketamine and the comparison groups at 5–7 days postpartum (SMD=−0.34, 95 % CI [−0.78, 0.09], $p = 0.12$, $I^2 = 80$ %).

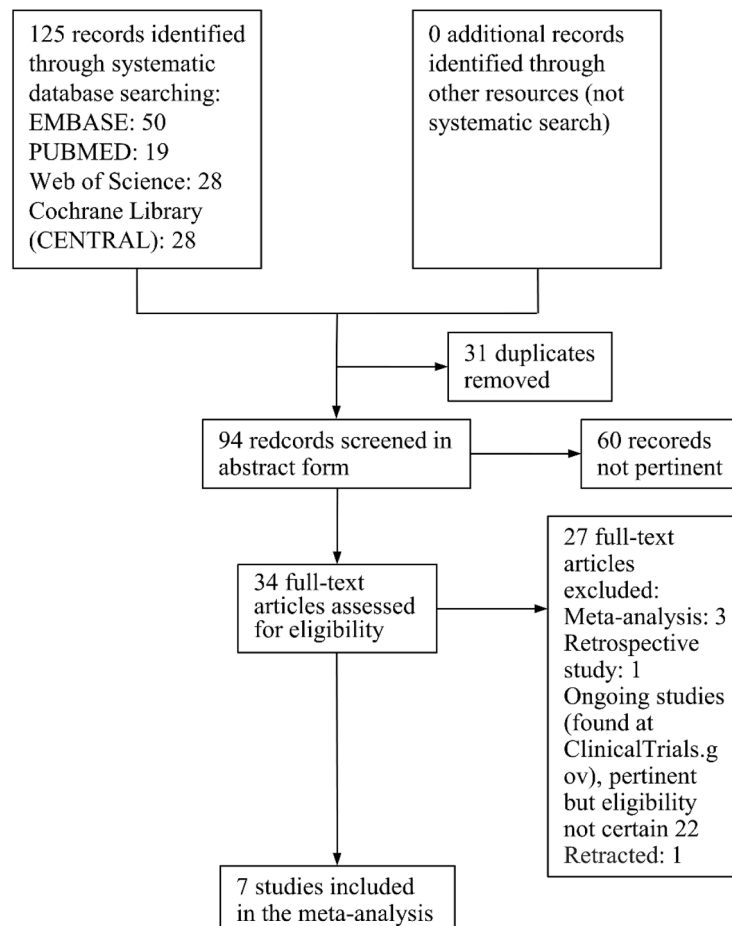


Fig. 1. Flow chart of study selection.

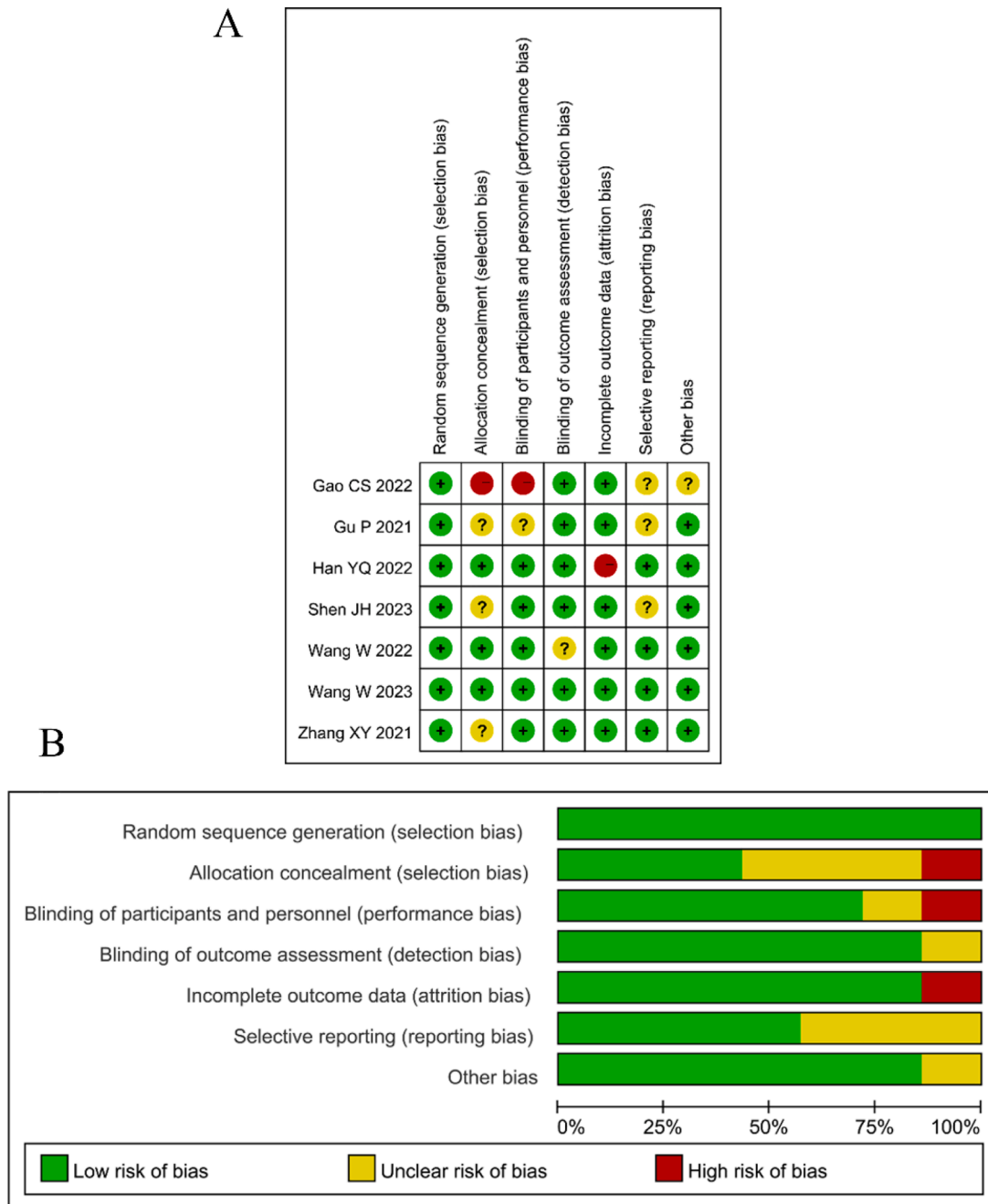


Fig. 2. Risk of bias assessment. A Risk of bias for individual studies. B The summary of the risk of bias.

3.4. Incidence of nausea and vomiting, dizziness, drowsiness and other adverse effects

Three studies reported the incidence of nausea and vomiting separately, and four studies did not distinguish between nausea and vomiting. There was no significant difference in the incidence of nausea and vomiting between the esketamine and comparison groups (RR=0.65, 95 % CI [0.34, 1.25], $p = 0.20$, $I^2 = 65$ %) (Fig. 5A). Subgroup analysis of the esketamine doses showed that the incidence of nausea and vomiting in the esketamine group was significantly lower than that in the comparison group in the low-dose subgroup (< 0.5 mg/kg) (RR=0.41, 95 % CI [0.17, 0.98], $p = 0.05$, $I^2 = 67$ %) (Figure S3). In the high-dose subgroup (≥ 0.5 mg/kg), there was no significant difference in the incidence of nausea and vomiting between the two groups (RR=1.30, 95 % CI [0.77, 2.20], $p = 0.32$, $I^2 = 0$ %) (Figure S3). The incidence of

postoperative nausea and vomiting was not affected by any administration methods (Figure S4). The incidence of dizziness (RR=3.35, 95 % CI [0.53, 21.42], $p = 0.20$, $I^2 = 84$ %) and drowsiness (RR=1.36, 95 % CI [0.68, 2.70], $p = 0.39$, $I^2 = 0$ %) was not significantly different between the esketamine and comparison groups (Fig. 5). One study reported that the Ramsay sedation scale (RSS) scores of the esketamine group were significantly higher than that of the comparison group, and there was no significant difference in the incidence of headaches (Figure S5). One study reported the incidence of hallucinations in the esketamine group was higher than that in the comparison group at 5 min after esketamine administration. Still, the difference was not statistically significant when leaving the operating room (Figure S6).

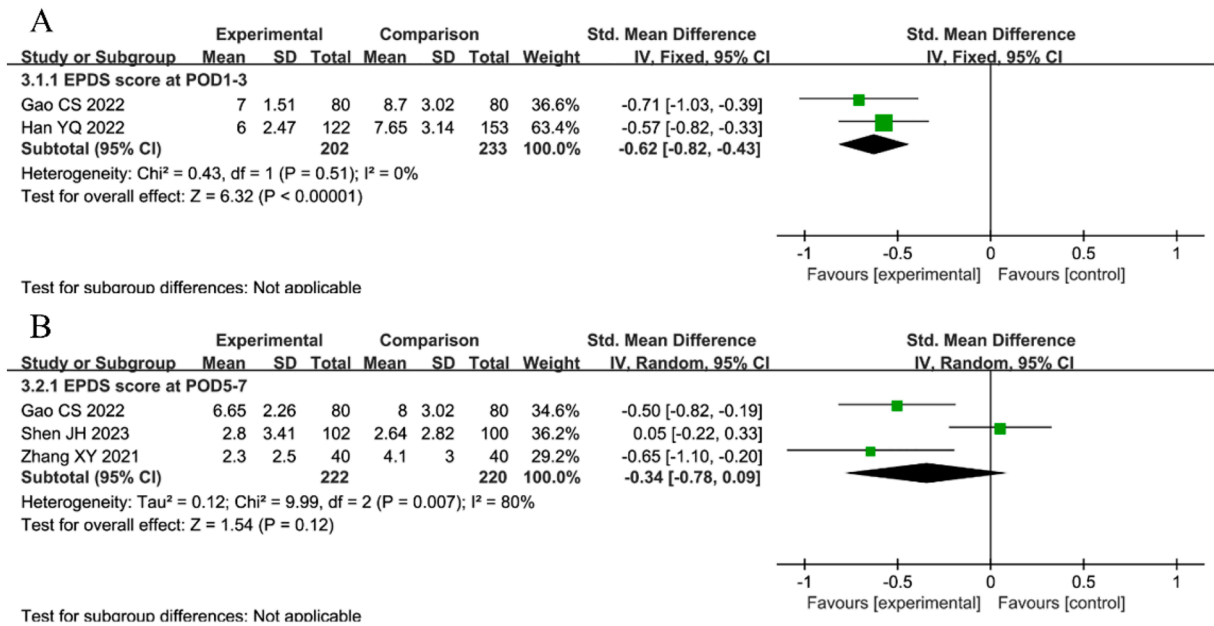


Fig. 3. The effects of perioperative administration of esketamine on postpartum EPDS scores. Forest plots of the postpartum EPDS scores in randomized controlled trials. POD postoperative day. CI confidence interval. df degrees of freedom.

3.5. Pain score 24 h and 48 h after surgery

Three studies reported postoperative pain scores at 24 h and 48 h. There was no significant difference in postoperative resting pain score between the esketamine group and the comparison group at 24 h ($SMD = -0.48$, 95 % CI $[-1.58, 0.62]$, $p = 0.39$, $I^2 = 97\%$) or 48 h ($SMD = -0.42$, 95 % CI $[-1.22, 0.39]$, $p = 0.31$, $I^2 = 95\%$) after surgery (Fig. 6). The two groups had no significant difference in coughing pain scores at 24 h ($SMD = -0.82$, 95 % CI $[-2.32, 0.68]$, $p = 0.29$, $I^2 = 96\%$) and 48 h ($SMD = -0.53$, 95 % CI $[-1.57, 0.52]$, $p = 0.32$, $I^2 = 93\%$) after surgery (Figure S7).

3.6. Subgroup analysis

To investigate whether moderating variables affect esketamine's ability to improve PPD incidence, subgroup analyses were performed based on esketamine administration methods (single infusion vs. continuous infusion administration of PCIA) (Figure S2) and esketamine doses (low dose vs. high dose) (Figure S1). The results were as follows (Table S2): both the high-dose group (≥ 0.5 mg/kg) ($RR = 0.48$, 95 % CI $[0.31, 0.74]$, $p = 0.001$, $I^2 = 0\%$) and the low-dose group (< 0.5 mg/kg) ($RR = 0.25$, 95 % CI $[0.14, 0.46]$, $p < 0.00001$, $I^2 = 62\%$) esketamine significantly reduced the prevalence of PPD within 5–7 days postpartum. Esketamine significantly reduced the prevalence of PPD 5–7 days postpartum in the PCIA ($RR = 0.30$, 95 % CI $[0.19, 0.49]$, $p < 0.00001$, $I^2 = 58\%$) and single infusion ($RR = 0.48$, 95 % CI $[0.29, 0.81]$, $p = 0.006$, $I^2 = 47\%$) subgroups. There was no significant difference between the esketamine and comparison groups.

3.7. Sensitivity analysis

Leave-one-out sensitivity analysis was performed to explore the stability and heterogeneity of the meta-analysis results. These results were stable and reliable. (Figure S8–S11).

3.8. Summary of findings for the main comparison

We used the principles of the GRADE system to assess the quality of evidence for the primary and secondary outcomes in this meta-analysis.

The quality of the evidence was mostly very low or low due to imprecision, risk of bias, and the heterogeneity of the seven studies included. The results are shown in Table S3.

4. Discussion

This meta-analysis found that esketamine could effectively reduce EPDS scores and the incidence of PPD after cesarean section. Even at 42 days postpartum, the incidence of PPD in the esketamine group was still significantly lower than in the comparison group. The prophylactic effect of esketamine on PPD was independent of the administration method and dose of esketamine. Esketamine did not increase the incidence of postoperative nausea and vomiting, dizziness, and drowsiness. The two groups had no significant difference in resting or coughing pain scores at 24 and 48 h after surgery.

Esketamine was effective in reducing the incidence of PPD after elective cesarean section, even at 42 days postpartum. Both human and animal studies have reported the antidepressant effects of ketamine and esketamine (Guo et al., 2023; Jiang et al., 2023; d'Andrea et al., 2023; Marwaha et al., 2023). A meta-analysis found that a single intravenous of ketamine during the perioperative period of cesarean section significantly reduced the PPD score and prevalence of PPD at one week postpartum. Still, there was no significant difference in PPD rate between the ketamine and comparison groups at four weeks postpartum (Li et al., 2022). Esketamine appears to be more effective than ketamine in long-term antidepressant effects, but further confirmation is needed. In addition, a meta-analysis showed that the intranasal esketamine combined with antidepressants was effective in reducing the severity of MDD (Hock et al., 2022; Papakostas et al., 2020). However, inconsistent results in our meta-analyses should not be ignored. Shen et al. found that a single intravenous infusion of 0.25 mg/kg of esketamine did not reduce the incidence of depression at 1, 2, or 4 weeks postpartum (Shen et al., 2023). Unfortunately, we did not find differences between this study and other studies in patients' characteristics, modes of administration and doses. In the study of Han et al. (Han et al., 2022), the lost to follow-up rate was over 20 %. We performed a sensitivity analysis of the number of lost to follow-up using the best-worst case analysis (Mavridis et al., 2014). Figure S12 showed the extremes of the incidence of PPD in the esketamine and comparison groups 1–3 days postpartum. The

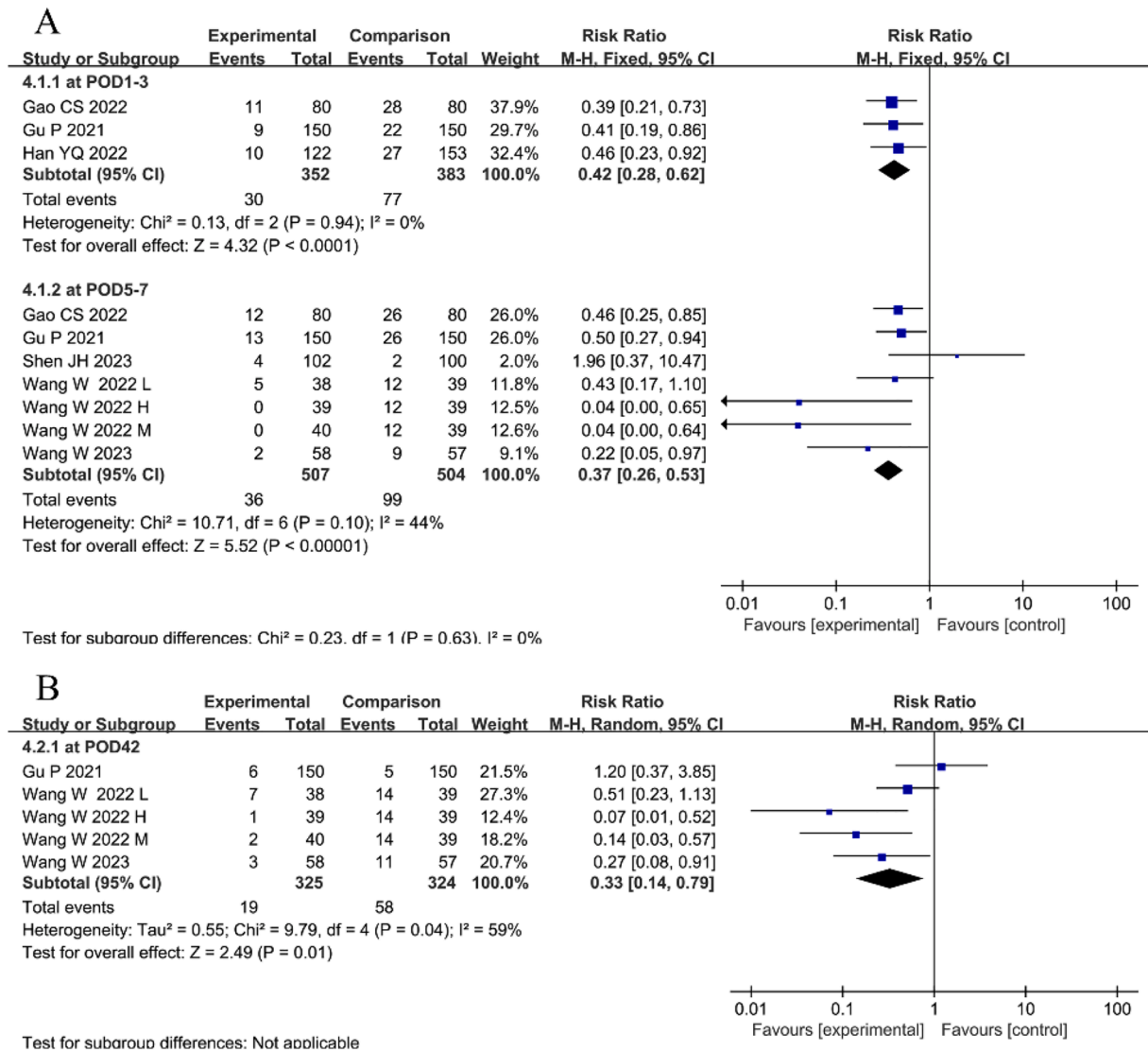


Fig. 4. The effects of perioperative administration of esketamine on the incidence of PPD. Forest plots of the PPD in randomized controlled trials. POD postoperative day. PPD Postpartum depression. CI confidence interval. df degrees of freedom.

pooled results of the complete case analysis, the extreme favouring esketamine group and the extreme favouring comparison group showed a strong beneficial effect of the esketamine group over the comparison group. Therefore, according to the above results, Han et al.'s study's high lost to follow-up rate had no significant effect on the pooled incidence of PPD at 1–3 days postpartum. More randomized controlled studies are needed to clarify the antidepressant efficacy of esketamine.

The mechanisms underlying esketamine's antidepressant effects have not been fully elucidated. Accumulating preclinical data suggest that brain-derived neurotrophic factor (BDNF) and mammalian target of rapamycin complex 1 (mTORC1) play a vital role in the antidepressant effects of esketamine (Salahudeen et al., 2020). In clinical studies, researchers found that the improvement in patients with postoperative depression after esketamine administration was associated with elevated serum levels of BDNF (Wang et al., 2020). Yang et al. demonstrated that the mTORC1 signalling pathway plays a role in the antidepressant effects of esketamine, but not (R)-ketamine (Yang et al., 2018). The rapid antidepressant effect of esketamine is possibly related to its effect on synaptic potentiation (Cornwell et al., 2012), which may enhance the relief of negative thinking and the prevention of negative spirals of depressed thoughts. The antidepressant effect of esketamine may be related to its immunomodulatory and anti-inflammatory effects,

as inflammatory cytokine levels are always elevated in patients undergoing surgery (Jiang et al., 2023). Serum interleukin-6 can be used as a predictive biomarker of the antidepressant effect of ketamine in patients with treatment-resistant depression (Yang et al., 2015). A recent study by our research group found that esketamine inhibited the production of pro-inflammatory cytokines IL-6, IL-1 β , and TNF- α , effectively improved postoperative depression-like symptoms in missed abortion patients (Jiang et al., 2023). Furthermore, esketamine is increasingly used as an analgesic for perioperative pain. There is a close relationship between depression and pain (Holmes et al., 2013). A meta-analysis reported that esketamine for postoperative analgesia after cesarean section found that postoperative pain was lower in the esketamine group than in the comparison group at 4 h, 12 h, 24 h, and 48 h after surgery (Liang et al., 2022). In this meta-analysis, esketamine had no significant effect on pain scores 24 or 48 h after surgery, in contrast to some previous studies. The possible reason is that only three studies in our research reported postoperative pain scores, and there was significant heterogeneity among the studies. In only one of the three studies, esketamine was observed to significantly reduce pain scores at 24 and 48 h after surgery (Zhang et al., 2021). Further studies are required to clarify whether esketamine can reduce postoperative pain. In addition to NMDA receptors, esketamine has been reported to interact with the μ - and κ -

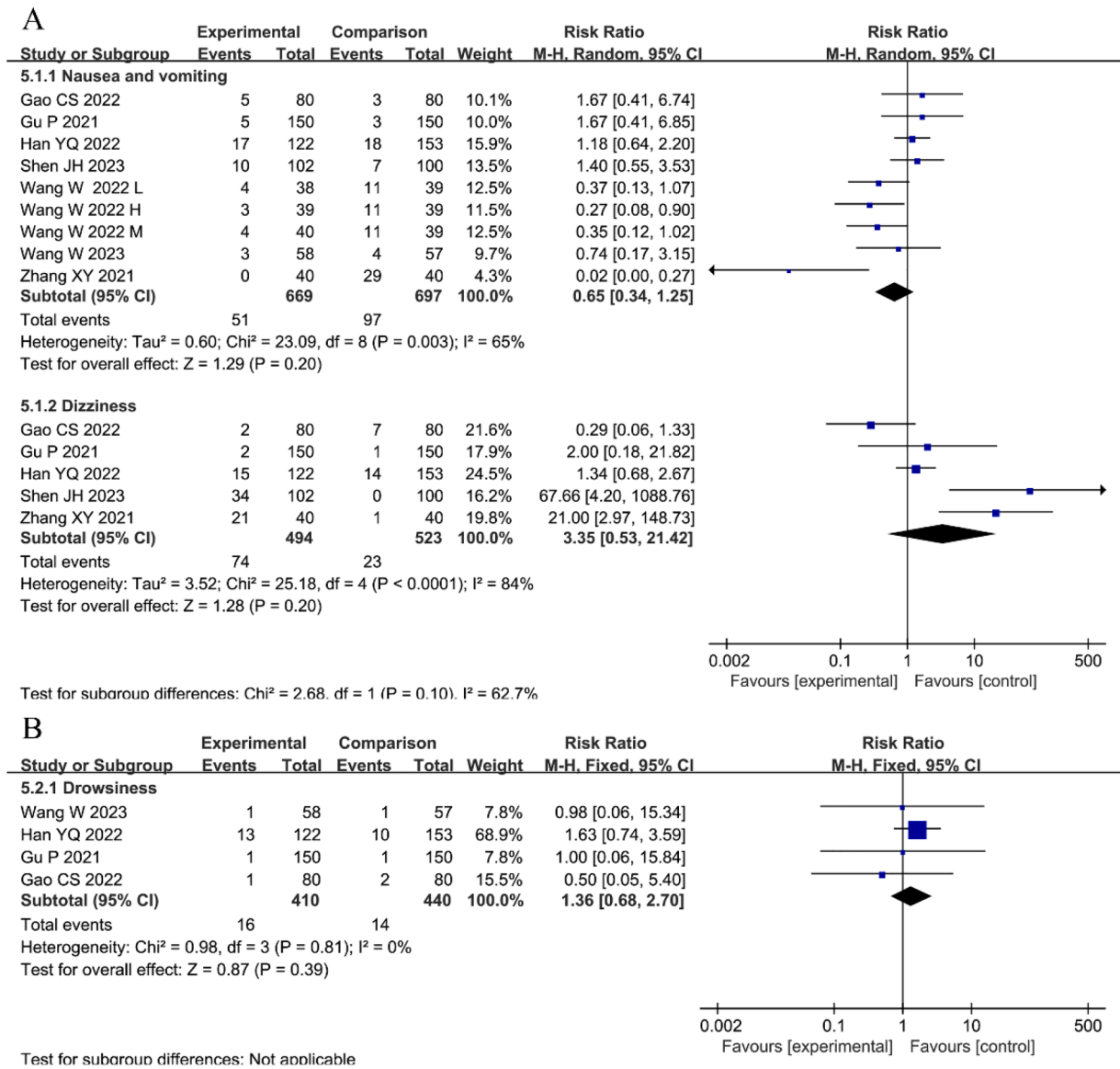


Fig. 5. Forest plots of adverse effects in randomized controlled trials. A Nausea and vomiting, and Dizziness. B Drowsiness. CI confidence interval, df degrees of freedom.

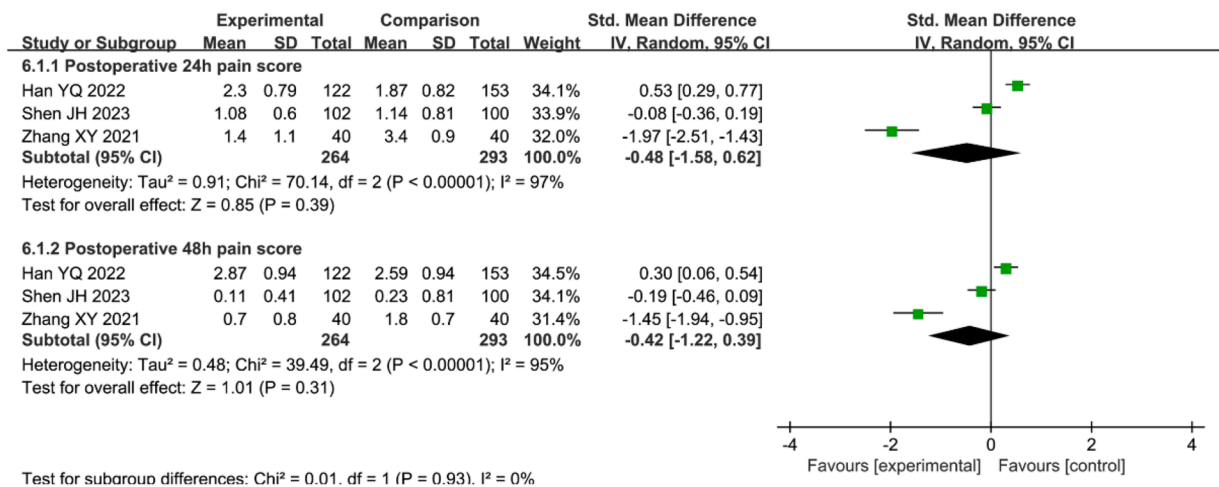


Fig. 6. The effects of perioperative administration of esketamine on postoperative pain scores. Forest plots of postoperative pain intensity in randomized controlled trials. CI confidence interval, df degrees of freedom.

receptors of the opioid system (Bonaventura et al., 2021). It was also found in clinical trials that the administration of the opioid antagonist naltrexone blocked the antidepressant effects of ketamine (Williams et al., 2019). Recently, however, Klein and colleagues (Klein et al., 2020) have reported that the opioid effects of racemic ketamine are necessary but not sufficient to exert its antidepressant effects. More clinical trials are needed to verify that esketamine's interaction with opioid receptors drives its rapid antidepressant effect. To investigate whether moderating variables affect esketamine's ability to improve PPD incidence, we performed a subgroup analysis based on the methods and doses of esketamine administration.

Almost all previous studies have shown that ketamine has a fast-acting antidepressant effect. However, the optimal dose of esketamine for PPD is still unknown. We analyzed the incidence of PPD at 5–7 days postpartum by subgroup analyses based on the esketamine dosage. Esketamine in both the high-dose (≥ 0.5 mg/kg) and the low-dose (< 0.5 mg/kg) groups significantly reduced the prevalence of PPD at 5–7 days postpartum. A recent double-blind, placebo-controlled study using different doses of esketamine (0.1, 0.2, 0.4 mg/kg, PCIA) demonstrated that 0.2 and 0.4 mg/kg of esketamine had better antidepressant efficacy than 0.1 mg/kg and control groups (Wang et al., 2022). In another study, it was demonstrated that high doses (0.5 and 1.0 mg/kg), not low doses (0.1 or 0.2 mg/kg) of ketamine, had antidepressant effects (Fava et al., 2020). The different results between the two studies may be due to differences in patients, drugs, and the modes of administration. Patients in this meta-analysis received esketamine at doses ranging from 0.1 to 1.0 mg/kg, with the most commonly used dose being 0.5 mg/kg. It is generally believed that adverse effects increase with increasing doses. Still, there was no significant difference in adverse effects between the high-dose (≥ 0.5 mg/kg) and low-dose (< 0.5 mg/kg) groups in the current meta-analysis. More randomized, double-blind studies are needed to determine the optimal dose of esketamine for preventing PPD.

Additionally, we conducted a subgroup analysis of esketamine administration methods to observe the safety of esketamine and its effect on the incidence of PPD. A single infusion of esketamine during surgery or continuous infusion after surgery can effectively prevent PPD, and the adverse effects associated with esketamine were not significantly different between the two subgroups. In a meta-analysis of Guo et al., continuous ketamine infusion was found to be effective in reducing ketamine-related adverse effects while maintaining its antidepressant effects compared to single-dose ketamine administration (Guo et al., 2023). The reasons for the differences may be that esketamine has fewer adverse effects than ketamine and the difference in the type of surgery and the type of anesthesia. A meta-analysis by McIntyre et al. compared the effects of different ketamine/esketamine administration routes (intravenous, intranasal, and oral) on mood disorders and found that the antidepressant efficacy of both intravenous ketamine and intranasal esketamine was supported by past studies (McIntyre et al., 2020). The effect size of intravenous ketamine was most significant at 2–6 days, while the effect size of intranasal esketamine was most significant at 24 h. In addition, the mean bioavailability of intranasal esketamine is estimated to be 48 % (Salahudeen et al., 2020), and its absorption also depends on the administration technique and the presence of nasal lesions. Therefore, compared with intranasal esketamine, intravenous esketamine has a higher bioavailability and is more convenient to control the dosage. Well-designed studies that directly compare intravenous esketamine with intranasal esketamine are currently missing and will be necessary in the future.

The safety of drugs is a matter of great concern in obstetrics. A recent clinical study published in the journal of JAMA Network Open examined the safety and efficacy of esketamine in elective cesarean section (Xu et al., 2023). A subanesthetic dose of esketamine (0.25 mg/kg) administered before surgery produced transient analgesic and sedative effects but did not induce significant neonatal inhibition. In addition, clinical studies have found that esketamine was relatively safe for infants and children (Pees et al., 2003). On the evidence so far, esketamine seems to

be safe and effective for use in cesarean section operations without affecting the outcome of newborns. In this study, we also found that esketamine did not increase the incidence of postoperative nausea and vomiting, dizziness, and drowsiness. Interestingly, postoperative nausea and vomiting incidence was significantly lower in the esketamine low-dose (< 0.5 mg/kg) subgroup, and this is similar to the results of a randomized multicenter clinical trial in patients undergoing laparoscopic appendectomy and cholecystectomy, in which the synergistic effect of esketamine and propofol anesthesia was found to reduce postoperative nausea and vomiting, and other short-term adverse events (Chen et al., 2023). Sedation and psychiatric symptoms such as hallucinations were common but only temporary and occur intraoperatively. Esketamine causes sympathetic excitation and dose-dependent blood pressure and heart rate increases. A subanesthetic dose of esketamine administered before surgery can effectively prevent maternal hypotension induced by spinal anesthesia (Orbach-Zinger et al., 2012; Zieleskiewicz et al., 2018). A single intravenous infusion of low-dose esketamine reduced the incidence of hypotension and eliminated tension in cesarean-section patients without increasing adverse events (Zhang et al., 2021). The increased blood pressure induced by esketamine is transient and usually disappears 2 h after administration (Zhang et al., 2021). Therefore, based on current evidence, esketamine appears to be safe and effective for perioperative obstetric patients because a single small dose administration can effectively prevent PPD without increasing the associated adverse effects.

The results of this meta-analysis should be interpreted with caution due to the following limitations, which are mostly related to the weaknesses of the original trials. (1) The total sample size was small and the heterogeneity of most results was high, possibly due to the small number of included studies and different administration methods and dosages. More RCT studies with large samples are needed to verify the results' reliability further. (2) The included studies covered two types of administration, including single infusion and continuous intravenous pump infusion. There are three timings of administration, including preincision, intraoperative and postoperative. Intraoperative refers to the period from delivery to 10 min after delivery. Although the dosages were all subanesthetic doses of esketamine, the dose range was still extensive, ranging from 0.1 to 1 mg/kg. (3) Among the outcome indicators, postoperative pain score and dizziness showed high heterogeneity in statistical analysis. However, due to limited data in the study, further subgroup analysis was not possible, affecting the accuracy of the results. (4) The studies included in this meta-analysis are all from China, which may bias the applicability of the findings in different countries. The reason for this situation may be related to the large base of PPD patients in China, and doctors are eager to find feasible solutions to alleviate the suffering of depressed patients. Intravenous esketamine was launched in China in 2019. Since then, studies on esketamine to alleviate perioperative and perinatal depression have increased in China. (5) The final limitation is the multiple comparisons from some studies, such as Wang et al., 2022, may have introduced the possibility of a type I error in the results.

5. Conclusion

The results of our meta-analysis have important implications for the effectiveness and safety of preventing PPD after elective cesarean section. Esketamine effectively reduced the EPDS scores and the incidence of PPD after cesarean section. Even at 42 days postpartum, the incidence of PPD in the esketamine group was still significantly lower than in the comparison group. The prophylactic effect of esketamine on PPD was independent of the administration methods and doses of esketamine. Low doses of esketamine can effectively reduce the incidence of postoperative nausea and vomiting. In the future, more studies with large samples and different countries are needed to confirm the credibility and validity of the results.

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Data availability

Correspondence and material requests should be addressed to Ming Jiang, Hongmei Yuan, or Shanwu Feng.

CRedit authorship contribution statement

Yazhou Wen: Writing – original draft, Methodology, Funding acquisition, Data curation. **Mingjie Mao:** Software, Methodology, Data curation. **Xian Wang:** Writing – review & editing, Methodology, Data curation. **Chenyang Xu:** Software, Data curation, Conceptualization. **Xueduo Shi:** Resources, Data curation, Conceptualization. **Ping Li:** Visualization, Data curation, Conceptualization. **Zijun Tian:** Formal analysis, Data curation, Conceptualization. **Ming Jiang:** Writing – review & editing, Supervision, Conceptualization. **Hongmei Yuan:** Writing – review & editing, Supervision, Conceptualization. **Shanwu Feng:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

All authors declare that they have no conflicts of interest.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.psychres.2024.115765](https://doi.org/10.1016/j.psychres.2024.115765).

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